Course Five

The Nuts and Bolts of Machine Learning

Instructions

Use this PACE strategy document to record decisions and reflections as you work through the end-of-course project. As a reminder, this document is a resource that you can reference in the future and a guide to help consider responses and reflections posed at various points throughout projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the questions in the Course 5 PACE strategy document
- Answer the questions in the Jupyter notebook project file
- Build a machine learning model
- Create an executive summary for team members and other stakeholders

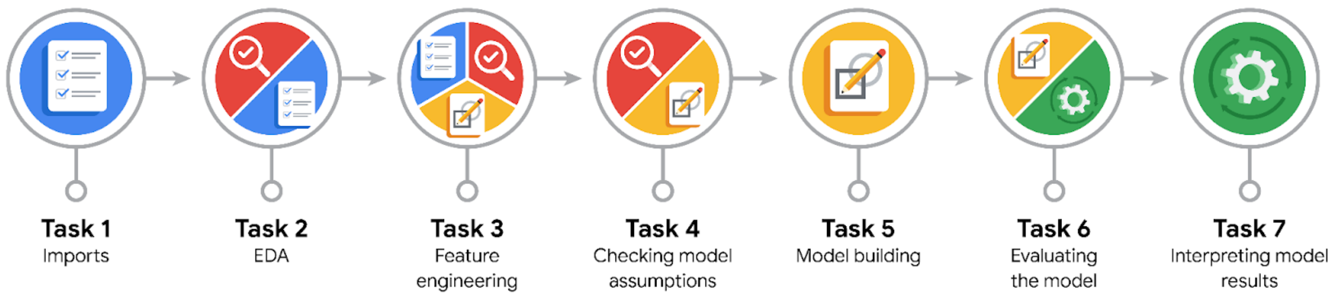
Relevant Interview Questions

Completing the end-of-course project will empower you to respond to the following interview topics:

- What kinds of business problems would be best addressed by supervised learning models?
- What requirements are needed to create effective supervised learning models?
- What does machine learning mean to you?
- How would you explain what machine learning algorithms do to a teammate who is new to the concept?
- How does gradient boosting work?

Reference Guide:

This project has seven tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- What are you trying to solve or accomplish?

We are tasked with building a random forest machine learning model to predict whether a customer's report is an opinion or a claim. TikTok receives a huge number of reports per day, and keeping up with these reports is time-consuming and requires a lot of effort; therefore, building a model that streamlines the classification process would save the team's time to address the real issues that impede the users' experience with the platform.

- Who are your external stakeholders that I will be presenting for this project?

The external stakeholders are the project manager (Mary Joanna), the Finance Lead (Margery Adebowale), and the Operation Lead (Maika Abadi) as well as Data Science Lead (Willow Jaffey) and Data Science Manager (Rosie Mae Bradshaw).

- What resources do you find yourself using as you complete this stage?

The resources that I used during this stage are key evaluation metrics (accuracy, precision, recall, and F1 score) and confusion matrix parameters (False negative, false positive, true positive, and true negative), with more emphasis on recall. That is because false negatives have a more computational and reputational risk impact on TikTok than false positives.

- Do you have any ethical considerations at this stage?

At this stage, it is important to keep in mind that the model's false negative predictions (mistakenly identifying claims regarding harmful, discriminative, or bullying videos were incorrectly identified as an opinion) would result in serious ethical considerations like violence, discrimination, bullying and so on that detrimentally affect user safety.

- Is my data reliable?

The data is highly reliable as it comes directly from TikTok. However, the target variable that we are trying to predict is binary, meaning we have to check class balance before building the model. This is because if we have more data that comes from one class, say more than 90%, then the model would predict the majority class, while ignoring the minority class, which leads to misidentifying the minority and failing to represent the real world problem they represent.

- What data do I need/would like to see in a perfect world to answer this question?

Cleaned, balanced, and representative data of different users from verified and unverified accounts, from different geographies, and from different age groups would be perfect for an accurate and thorough analysis.

- What data do I have/can I get?

We have a dataset containing **19,382 rows** and **12 columns**.

Predictive Features (10): `claim_status`, `video_duration_sec`, `video_transcription_text`, `verified_status`, `author_ban_status`, `video_view_count`, `video_like_count`, `video_share_count`, `video_download_count`, and `video_comment_count`.

Administrative Columns (2): Row ID (`#`) and `video_id` (used for identification, not prediction).

- What metric should I use to evaluate success of my business/organizational objective? Why?

The metric that will be used to evaluate the success of the model is recall because recall focuses on identifying all actual claims. In this case, a false negative (where the model incorrectly predicts a report as being an opinion while it is a claim) is more costly than a false positive (where the model misidentifies an opinion as a claim). While false positives may require unnecessary review, it does not harm the end-user the way the missing claim does. Additionally, since the main purpose of the model is to increase response time and system efficiency, a delay in responding to users' claims can detrimentally affect their overall experience and lead to dissatisfaction.

PACE: Analyze Stage

- Revisit "What am I trying to solve?"Does it still work? Does the plan need revising?

Building a model to automatically classify videos as being opinion or claim to meet the overall objective of increasing response time and system efficiency.

In the analysis phase, data exploration and cleaning are implemented to prepare the data for feature engineering, which will be conducted during the construction phase. Such steps are very crucial to increase a model's performance efficiency and accuracy.

Yes, it still works. We have a column "claim_status", which is the target variable that stores whether a video is an opinion or claim, and video features that will be selected from and converted, if categorical, into numeric during feature engineering.

No, the main plan is still the same, but the given dataset needs preparation and manipulation. This means we will drop unnecessary columns, such as the row number and video_id, and transform categorical variables into numeric values, as " RandomForestClassifier()" does not accept any string values.

- Does the data break the assumptions of the model? Is that ok, or unacceptable?

Since we are building a random forest model, which is a non-parametric model, we do not have to worry about strict assumptions, such as linearity, normality, and homoscedasticity. That means while the analysis shows that there are some outliers, the random forest model could handle them very well because it uses many decision trees; each tree splits the data based on rules rather than mathematical distance and then takes the average or the voting majority, which minimizes the effect of extreme outliers.

- Why did you select the X variables you did?

I selected the X variables that would have a direct effect in determining whether a video is an opinion or claim. Columns that store the row number and the one that stores the video ID would not add any value, other than consuming computational resources as we tune the hyperparameters and build the model.

- What are some purposes of EDA before constructing a model?

The purpose of EDA is to explore, check for null values, duplication, and outliers, and determine the target variable and feature variables.

- What has the EDA told you?

The EDA told me that the "claim-status" would be our target variable that we want the model to predict. Additionally, there are columns that need to be dropped and others that need to be converted into numeric values before building the model.

- What resources do you find yourself using as you complete this stage?

I used pandas functions to explore and manipulate the data and check class balance.

PACE: Construct Stage

- Do I notice anything odd? Is it a problem? Can it be fixed? If so, how?

The odd thing is that the precision and recall scores are near perfect, especially for the XGBoost model. Such high scores are not usual in the real world of data science and machine learning, which may indicate high correlation between the target variable and the features or even data leakage (which occurs when the answer is inadvertently included while the model is still in the training process). It could also mean that the data is synthetic and does not apply to real-world situations.

A high score may indicate that the model is overfitted, meaning it performs very well on the training data but not so well on a new dataset, which leads to inaccurate predictions and misleading results.

Yes, it could be solved by dropping post-publication features (view count, like count, comment count, and download count) that lead to data leakage. These engagement metrics are usually included after the video is posted, and using them to predict whether a video is a claim or opinion means that the prediction would not take place until the video has gone live and gathered those stats.

- Which independent variables did you choose for the model, and why?

The independent variables that I chose for the model are “video_duration_sec " “video_transcription_text " (after being converted into numeric values), “verified_status" and "author_ban_status”. I chose them because they are available right when the video is posted, which prevents data leakage and leads to accurate predictions. I intentionally dropped post-publication video engagement metrics as I noticed that they lead to unusually high evaluation metric scores, which connotes an overfitted model.

- How well does your model fit the data? What is my model’s validation score?

It is not perfect, but still good. What really matters is that the model is not overfitted and could work well with new datasets. The validation score for the random forest is 0.80 and for the XGBoost 0.71.

- Can you improve it? Is there anything you would change about the model?

Yes, it could be improved if we include a wider range for each hyperparameter during tuning, but that would come with the cost of more computational resources and time. We could also explore and provide more relevant features (I.e., audio features, follower count, video content, etc).

Since we dropped the features that potentially could lead to an overfitted model, the model is now good, and I would not change anything other than improving its performance.

- What resources do you find yourself using as you complete this stage?

I used various libraries, such as pandas for data analysis and manipulation, seaborn and matplotlib for visualization, and sklearn for importing models like the RandomForestClassifier and XGBClassifier. Additionally, I used the evaluation metrics to examine how each model performs, what the score is, and what the parameters are.

PACE: Execute Stage

- What key insights emerged from your model(s)? Can you explain my model?

past The most predictive features are video transcription text length, followed by the author-ban-status-active. The results make sense as claims usually require more explanation, evidence, and context, which translates into longer transcriptions. The authors who report claims are most likely to be heavily scrutinized and placed under banned or under review status, making active status highly predictive, as it strongly signals the absence of a claim. The absence of the claim here is valuable as it allows the model to reveal patterns in the data

- What are the criteria for model selection?

The main criteria used for model selection is recall score as the main goal is to predict as many claims as possible. That is because missing an opinion would not be as costly as missing a claim. Beside recall, other common criteria used for model selection are precision, f1 score, accuracy, depending on the business objective. Sometimes a recall score is prioritized, other times a precision score would be the main criteria, and in other scenarios f1 would be the target score. Also, explainability and computational efficiency are among the most important criteria as they help in understanding how the model makes predictions and how quickly and economically the model runs.

- Does my model make sense? Are my final results acceptable?

As seen in the classification report, the model yielded a precision score of `0.67, recall score of 0.76, and f1 score of 0.72. So,` the model makes sense and the final results are acceptable, though there is still room for improvement as more data is collected.

- Do you think your model could be improved? Why or why not? How?

"Yes, it could be improved by collecting more data and extending hyperparameter ranges to test different combinations.

- Were there any features that were not important at all? What if you take them out?

Post-publication features, such as video view count, video like count, video share count, video download count, video comment count, were dropped to prevent data leakage. Because these metrics are highly correlated with the outcome and would not realistically be available until after a video is published and receives engagement, including them while training the model gives it the answer key, which leads to an absolutely overfitted model.

- What business/organizational recommendations do you propose based on the models built?

The random forest model yields more accurate predictions than XGBoost, therefore I recommend deploying it for future predictions. Additionally, further analysis and feature extraction must be implemented, especially for video transcription text length. That is because the model identified it as the key feature in predicting claim status. We can use n-grams to extract common words and sentiment analysis on videos flagged as claims to reveal the underlying shared patterns among claims.

- Given what you know about the data and the models you were using, what other questions could you address for the team?

What is the average video transcription text length across claims vs opinions ?

How do post publication metrics (the number of likes, comments, shares, views, and downloads) differ between verified and unverified accounts?

The number of videos posted from active, under review, and banned accounts and their post publication metrics?

- What resources do you find yourself using as you complete this stage?

In this phase, various libraries were used. This includes pandas for operation, matplotlib for visualization, and sklearn for model construction, evaluation, and hyperparameters tuning.

- Is my model ethical?

Yes, the model is ethical, as the data that was used to train and test it is neutral and does not represent any group over the other. The target variable is also balanced, meaning both classes (claim and opinion) are equally represented with roughly 50/50 split.

- When my model makes a mistake, what is happening? How does that translate to my use case?

When the model makes a false positive mistake (incorrectly identifies an opinion as a claim), at worst, human moderators would spend more time reviewing it. However, when the model makes a false negative mistake, users' safety and the overall platform integrity are put at risk. This exactly explains why we prioritize recall score over precision to catch as many claims as possible.